School of Technology and Engineering (SOTE)

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Computing Group Project

# UAT Test Plan


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## 1.0 Scope

### 1.1 Objectives & Business Requirements

The primary objective of User Acceptance Testing (UAT) is to validate that the NFCampus prototype meets its business goals and user requirements before final delivery. This testing phase focuses on confirming that the system provides a reliable backup authentication mechanism for campus users when their physical access cards are unavailable.

Key business requirements to be validated:

- Reliability: The system must provide a measurable success rate in triggering access authorization signals when a smartphone is tapped on the custom NFC reader.

- Usability: The process of registering a new account and linking a card must be intuitive and efficient for users, leveraging OCR to minimize manual data entry.

- Security: The “One Device, One Account” policy must be strictly enforced. This is to prevent multiple accounts on a single device or a single card from being linked to multiple users.

- Functionality: The core fallback mechanism (Phone to Custom Reader) must be fully functional, providing clear feedback such as haptic and visual to the user for both successful and failed access attempts.

### 1.2 Scope of the Project

The UAT will cover end-to-end user journey within the controlled environment that are defined in the project scope. This includes all core features of the Android application and its interaction with the custom-built ESP32 / PN532 NFC reader.

The testing will include:

- User registration and email verification flow
- OCR-based data extraction from physical student cards
- NFC card UID captures and links to a user account
- The “One Device, One Account” enforcement
- NFC-based access authentication (HCE) with the custom reader

- Viewing activity logs
- Account termination and data cleanup

The testing will exclude:

- Integration with the university's existing production access control readers
- Performance on iOS devices
- Full-scale campus deployment or load testing
- Advanced cryptographic security features

## 2.0 Testing Team

(Table 2.0 Testing Team)

| Role | Name | Responsibility |
|---|---|---|
| UAT Lead | Eu Jin Yang | Oversees the UAT process, coordinates with the development team, and resolves critical issues. |
| Test Coordinator | Teoh Yi Shan | Schedules test sessions, manages test data, and communicates with stakeholders. |
| Technical Analyst | Ng Kean Chun | Assists in setting up the test environment, monitors system logs, and verifies technical accuracy. |
| End-User Tester | Yong Ken | Executes test cases as an end-user, providing feedback on usability and functionality. |
| Additional Testers | Volunteers (Chan Jie Xin) | A student from logistics programme that is not a part of the development team to perform unbiased testing. |

## 3.0 Milestones & Deliverables

### 3.1 Design & Testing Process

- Status: Complete. UAT Plan drafted based on the finalized system requirements and prototype.
- Deliverable: This UAT Test Plan document.

### 3.2 Staging Environment

A dedicated, stable environment for UAT.

- Status: To be set up. This will consist of:
    - The production version of the Android app (Signed APK) is installed on test devices.
    - The Firebase project is in a dedicated testing mode.
    - The custom ESP32 / PN532 NFC reader and LCD display, configured and ready.
- Deliverable: A stable staging environment ready for testing.

### 3.3 Training

A brief orientation for the testers on the system's purpose and how to execute the test cases.

- Status: Pending. A short training session will be conducted by the UAT Lead.
- Deliverable: Completed training session and testers ready to begin.

### 3.4 UAT Execution

The period during which testers will execute the test cases, record results, and report any issues or bugs.

- Status: Pending
- Deliverable: Completed test case execution logs and a list of reported issues.

### 3.5 Reporting & Data Analysis

The UAT Lead will compile the results, analyze the success rate of test cases, and create a summary report for stakeholders.

Status: Pending

Deliverable: UAT Summary Report, including pass / fail statistics, a list of critical bugs, and a final recommendation.

## 4.0 Environmental Requirements

### 4.1 Hardware Requirements

(Table 4.0: Hardware Requirements)

| Item | Specification | Quantity |
|---|---|---|
| Android Phone | Android 8.0 (Oreo) or higher, with NFC and HCE support, active internet connection (Wi-Fi / Mobile Data). | 3 – 4 |
| Campus Access Card | A legitimate campus physical RFID / NFC card to simulate the “original” card. | 2 – 3 |
| Custom NFC Reader | The assembled NodeMCU ESP32 microcontroller / PN532 prototype with LCD display. | 1 |
| PC / Laptop | For accessing the Firebase console to monitor logs. | 1 |

### 4.2 Software Requirements

(Table 4.1: Software Requirements)

| Item | Version / Specifications |
|---|---|
| NFCampus Android App | Latest release build. |
| Firebase Console | Access to the project’s Firestore database and Authentication. |
| Node.js Server | Running locally to host the web-based verification page. |
| Web Browser | Latest version of Chrome, Firefox, or Edge. |

## 5.0 Features to be Tested

### 5.1 Feature 1: User Registration & Email Verification

5.1.1 Pass / Fail Criteria

- Pass: A new user can successfully register, verify their email via a link, and be guided to the card linking process. All data is correctly stored in Firebase. The system prevents registration with an already used email address.

5.1.2 Test Cases

(Table 5.0: Test Cases for Feature 1)

| Test Case ID | Test Scenario | Test Steps | Expected Result | Actual Result | Status (Pass / Fail) |
|---|---|---|---|---|---|
| TC-01-01 | Successful Registration | 1. Open app.<br>2. Click “Register here”.<br>3. Enter a valid, unused email and password.<br>4. Click “Register”. | A verification email is sent. User is informed to check their email. | Email sent to testuser@example.com. “Verification email sent” message displayed. | Pass |
| TC-01-02 | Email Verification | 1. After registration, open the verification link in the email.<br>2. Return to the app. | User is automatically notified of verification and prompted to log in. | After clicking link, browser shows “Email verified”. App shows login screen with success toast. | Pass |
| TC-01-03 | Duplicate Email Registration | 1. Attempt to register with an email that is already in the system. | An error message is displayed: “Email address is already in use.” | “The email address is already in use by another account.” error appears. | Pass |

### 5.2 Feature 2: Card Linking via OCR & NFC

5.2.1 Pass / Fail Criteria

- Pass: The user can scan their physical student card via OCR, review the data, and then capture the physical card’s UID via the phone’s NFC. The data is successfully stored in Firestore and SharedPreferences.

5.2.2 Test Cases

(Table 5.1: Test Cases for Feature 2)

| Test Case ID | Test Scenario | Test Steps | Expected Result | Actual Result | Status (Pass / Fail) |
|---|---|---|---|---|---|
| TC-02-01 | OCR Data Extraction | 1. Log in with a verified account.<br>2. Navigate to card setup.<br>3. Tap “Scan Front / Back of Card”.<br>4. Take a clear picture of the card. | The app extracts and displays the student’s name, ID, and course for user confirmation. | Extracted text: “Eu Jin Yang”, “BSSE2506021”, “Computer Science”. Preview screen shows correct data. | Pass |
| TC-02-02 | UID Capture | 1. After confirming OCR data, place the physical campus card on the back of the phone.<br>2. Tap “Start”. | The app reads the card’s UID and displays “Card Linked Successfully”. | UID 01:02:03:04 captured. Success message displayed. Firestore document updated with cardUID. | Pass |
| TC-02-03 | “One Device, One Account” | 1. After linking a card to User A, attempt to log in with a different account (User B) on the same device. | The login is rejected with an error message about the device being bound to another user. | “This device is already linked to another account. Please use the correct device.” error appears. | Pass |

### 5.3 Feature 3: NFC-Based Access Authentication

5.3.1 Pass / Fail Criteria

- Pass: With the app open and a valid UID stored, tapping the phone on the custom reader grants access. Tapping with an invalid / absent UID denies access. The LCD display and phone haptics provide correct feedback within 500ms.

5.3.2 Test Cases

(Table 5.2: Test Cases for Feature 3)

| Test Case ID | Test Scenario | Test Steps | Expected Result | Actual Result | Status (Pass / Fail) |
|---|---|---|---|---|---|
| TC-03-01 | Grant Access | 1. Log in with a user account that has a linked card.<br>2. Open the main dashboard.<br>3. Tap the phone on the custom NFC reader. | - Reader: LCD displays “Access Granted”.<br>- Phone: Haptic feedback (short vibration).<br>- Access log is created in Firestore. | Reader LCD: “Access Granted”. Phone vibrated once. Firestore activity_logs shows entry with granted: true | Pass |

| | | | | | |
|---|---|---|---|---|---|
| TC-03-02 | Deny Access | 1. Log in with a user account that has no linked card.<br>2. Tap the phone on the custom NFC reader. | - Reader: LCD displays “Access Denied”.<br>- Phone: Haptic feedback (longer vibration / pattern for failure).<br>- No log entry? (As per spec, reader doesn’t log to cloud). | Reader LCD: “Access Denied”. No new log in Firestore. | Pass |
| TC-03-03 | App in Background | 1. Log in with a valid account.<br>2. Put the app in the background.<br>3. Tap the phone on the reader. | Access is not granted (as HCE is only active when the app is in the foreground). | Reader LCD: “Access Denied”. Phone no haptic feedback. | Pass |

### 5.4 Feature 4: Account Management & Security

5.4.1 Pass / Fail Criteria

- Pass: Users can view their access history, change their email/password, and terminate their account. Upon termination, all associated data is atomically deleted from Firestore, and local credentials are wiped.

5.4.2 Test Cases

(Table 5.3: Test Cases for Feature 4)

| Test Case ID | Test Scenario | Test Steps | Expected Result | Actual Result | Status (Pass / Fail) |
|---|---|---|---|---|---|
| TC-04-01 | View Access History | 1. Log in to a user account.<br>2. Navigate to “Home” section and scan the QR code.<br>3. Verify the logs shown. | The screen displays a list of recent access attempts (logins, etc.) for that specific user. | Shows recent entries with timestamps and outcomes. Matches Firestore data. | Pass |
| TC-04-02 | Change Email | 1. Log in.<br>2. Navigate to settings and choose “Change Email”.<br>3. Enter a new email. | A verification email is sent to the new address. The account email is updated after verification. | Email sent to new address. After verification, login with new email works. Old email no longer accepted. | Pass |

| | | | | | |
|---|---|---|---|---|---|
| TC-04-03 | Account Termination | 1. Log in.<br>2. Navigate to settings and choose “Terminate Linked Card”.<br>3. Confirm the action. | - User is logged out.<br>- On Firestore, the user document and all nested activity_logs are deleted.<br>- On the phone, SharedPreferences are cleared. | User logged out. Firestore shows user document removed. Reopening app shows no stored credentials. | Pass |